DATA ANALYTICS PROJECT

Market Basket Analysis

Identifying cross-sell and bundle opportunities in retail transaction data

Anthony Maraka | MySQL | 4,845 baskets, 3,255 products

Dataset and scope

4,845
baskets of 2–50 items

3,255
distinct products

408
mean basket value (median 300)

17.2
mean items per basket (median 15)

Cleaning rules

- Cancelled invoices, non-positive quantity or price removed
- Non-product codes and carriage removed
- One line per product per basket

From raw data to baskets

- 164,693 raw transaction rows
- 83,158 product lines after cleaning
- 4,845 baskets, the denominator for support

Source: Online Retail dataset, 541,909 transaction lines

Revenue sits mostly in the long tail

Top 5 products	Baskets	Share
White Hanging Heart T-Light Holder	483	10.0%
Regency Cakestand 3 Tier	423	8.7%
Jumbo Bag Red Retrospot	417	8.6%
Assorted Colour Bird Ornament	368	7.6%
Party Bunting	367	7.6%

36.0%

of revenue comes from the 144 products (4.4%) found in 100+ baskets

64%

of revenue sits in the 3,111 rarer products

Shares are of 2–50 item baskets (n = 4,845)

From 1,444 pairs to a shortlist of 440

1,444

Pairs kept

In at least 20 baskets

760

Cross-family pairs

Different products, not variants

453

Meaningful lift

Lift of at least 3

440

Shortlisted

On sale together 90+ days and still selling

10

Ranked list

Top pairs by revenue, outlier-screened

Metrics used

Support
Share of baskets containing both products

Confidence
Share of buyers of A who also buy B

Lift
Above 1 means bought together more than chance

All thresholds are set in a single parameters table

Frequent pairs are mostly product variants

88 of 100

of the most frequent pairs are variants of one product, such as different designs of a jumbo bag

Variants vs cross-sell

Pair type	Pairs	Avg lift
Same family	684	21.5
Cross family	760	7.1

A high lift is a real signal. Among all possible pairs of popular products, only 61.5% have lift above 1 and 36.1% have lift below 1.

Same family = pairs sharing 2 or more meaningful product-name words

Recommended bundle candidates

Pair (passed the outlier check)	Baskets	Support	Lift	Pair revenue
Regency Cakestand 3 Tier + Roses Regency Teacup and Saucer	92	1.90%	5.3	9,973
Spotty Bunting + Party Bunting	99	2.04%	5.7	9,322
Regency Cakestand 3 Tier + Green Regency Teacup and Saucer	81	1.67%	5.5	8,408
Jumbo Shopper Vintage Red Paisley + Jumbo Bag Red Retrospot	115	2.37%	6.7	6,706
Regency Cakestand 3 Tier + Pink Regency Teacup and Saucer	60	1.24%	5.5	6,319

The cakestand is the anchor

47% of teacup buyers also buy the cakestand, but only 14–22% of cakestand buyers add a given teacup colour.

Five pairs excluded

Four doormat pairs and one heart pair fail the outlier check: one basket makes up 39–59% of their revenue.

Pair revenue = revenue of both products in baskets containing the pair

Validation, limitations and next steps

Validation

- Automated checks recount baskets, pairs, support, confidence and lift
- Final list matches independent recounts from the cleaned data
- Outlier screen flags pairs where one basket exceeds 20% of revenue

Limitations

- Margin of 40% is assumed: the data has no cost column
- Association, not causation: pair baskets are larger (745 vs 408)
- Only baskets of 2–50 items; family labels use shared words

Next steps

- A/B test a cakestand and teacup offer
- Replace the assumed margin with real cost data
- Add a second ranking by support and lift